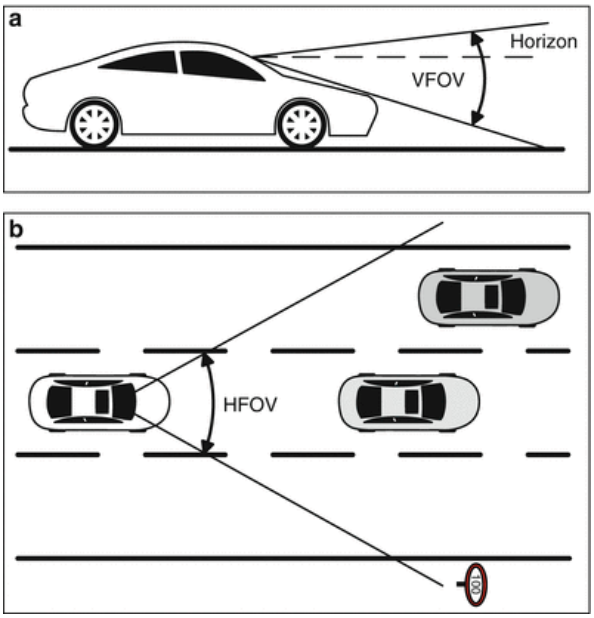
Computer Vision
Lecture 6: Camera Structure

Yeungnam University
MSP lab.
Ho-Youl JUNG

영남대학교 Yeungnam University

FoV(Field of View) 시야각

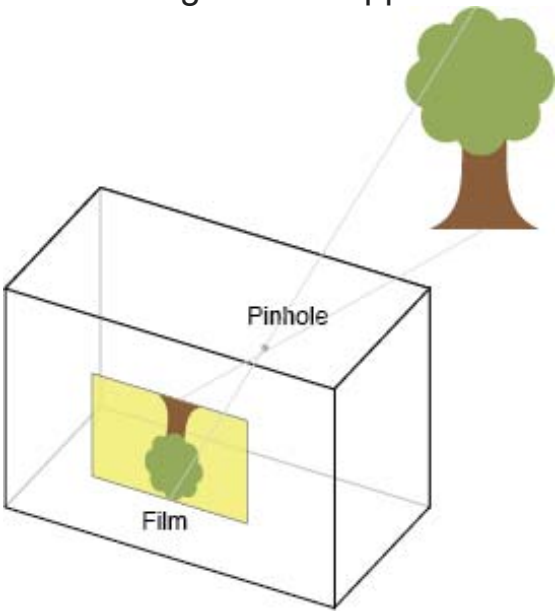
- Vertical FoV
- Horizontal FoV

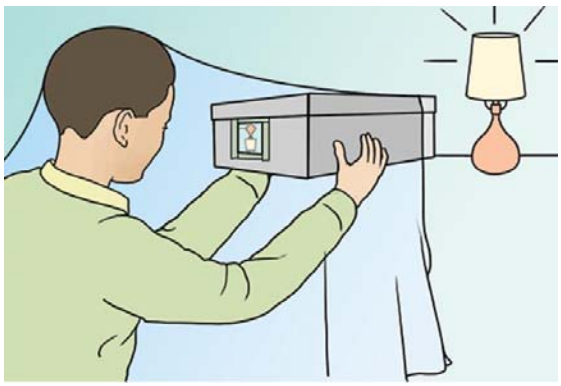




Pinhole Camera

- **Pinhole camera** is a simple camera without a lens but with a tiny aperture (pinhole)
- Light from a scene passes through the aperture and projects an inverted image on the opposite side of the box

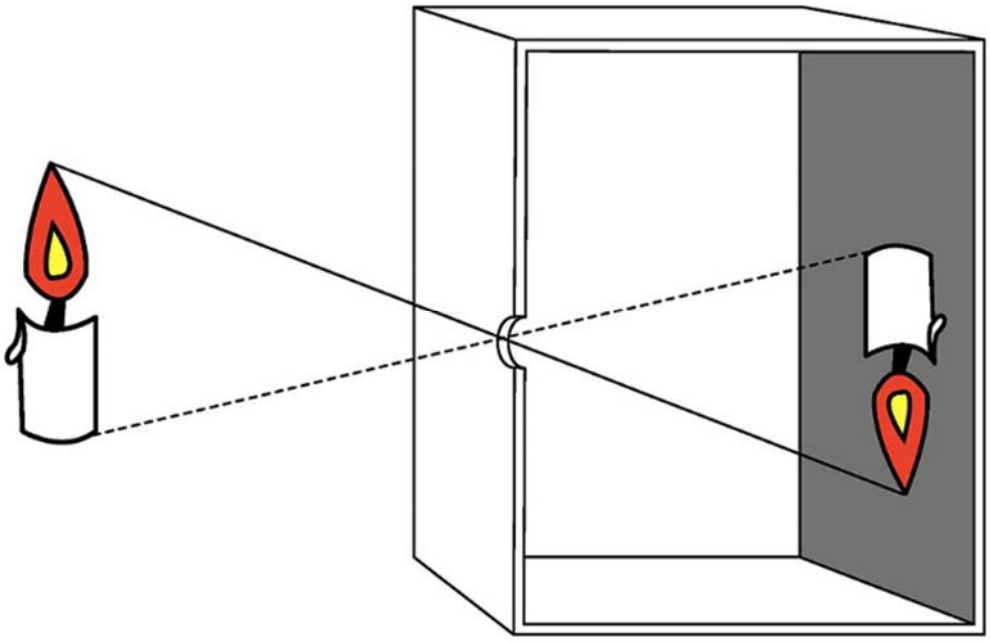




https://kids.nationalgeographic.com/explore/books/pinhole-camera/

https://ksimek.github.io/pinhole_camera_diagram/

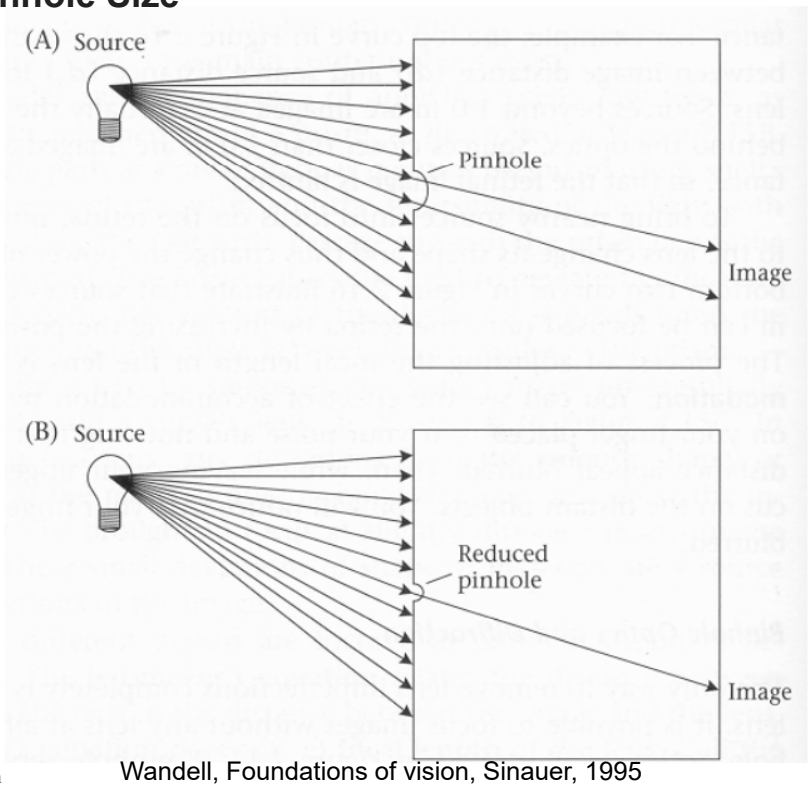
Pinhole camera



https://en.wiktionary.org/wiki/pinhole_camera

Pinhole Camera

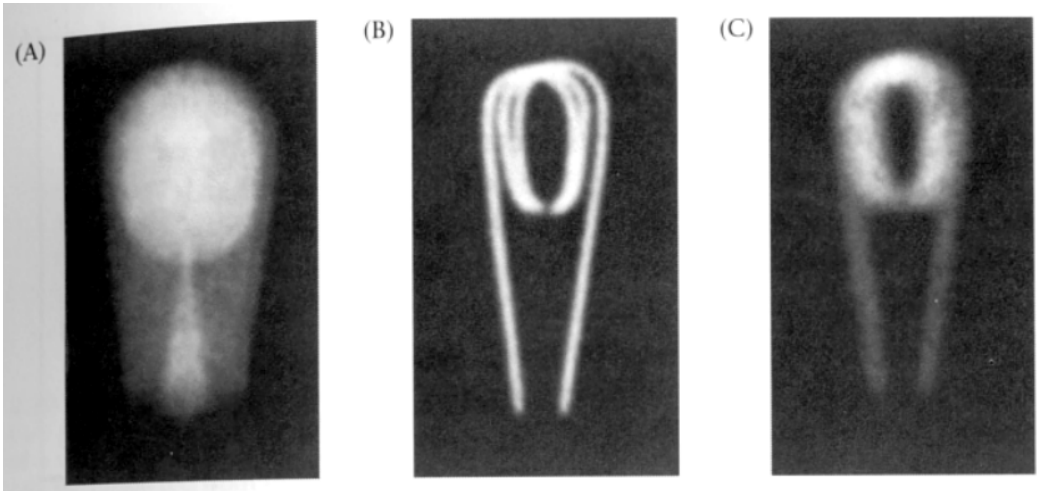
- **Effect of pinhole Size**



Wandell, Foundations of vision, Sinauer, 1995
http://6.869.csail.mit.edu/fa13/lectures/lecture10imageformation.pdf

Pinhole Camera

- **Effect of pinhole Size**

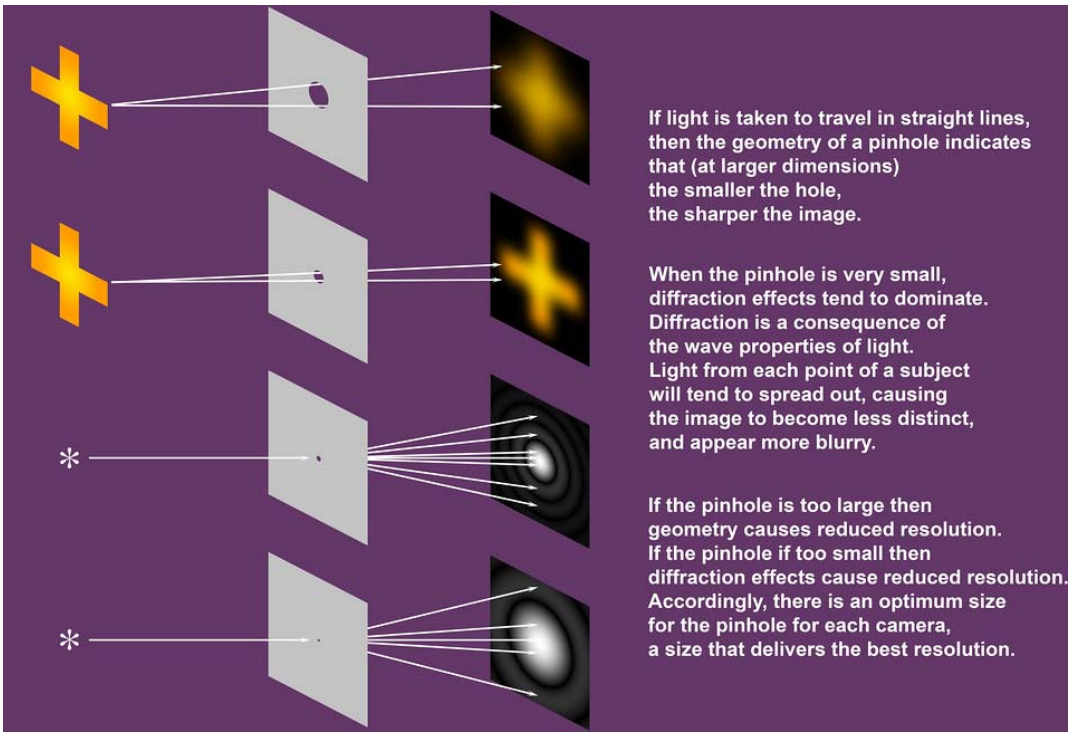


2.18 DIFFRACTION LIMITS THE QUALITY OF PINHOLE OPTICS. These three images of a bulb filament were made using pinholes with decreasing size. (A) When the pinhole is relatively large, the image rays are not properly converged, and the image is blurred. (B) Reducing the size of the pinhole improves the focus. (C) Reducing the size of the pinhole further worsens the focus, due to diffraction. From Ruechardt, 1958.

Wandell, Foundations of vision, Sinauer, 1995
http://6.869.csail.mit.edu/fa13/lectures/lecture10imageformation.pdf

Pinhole Camera

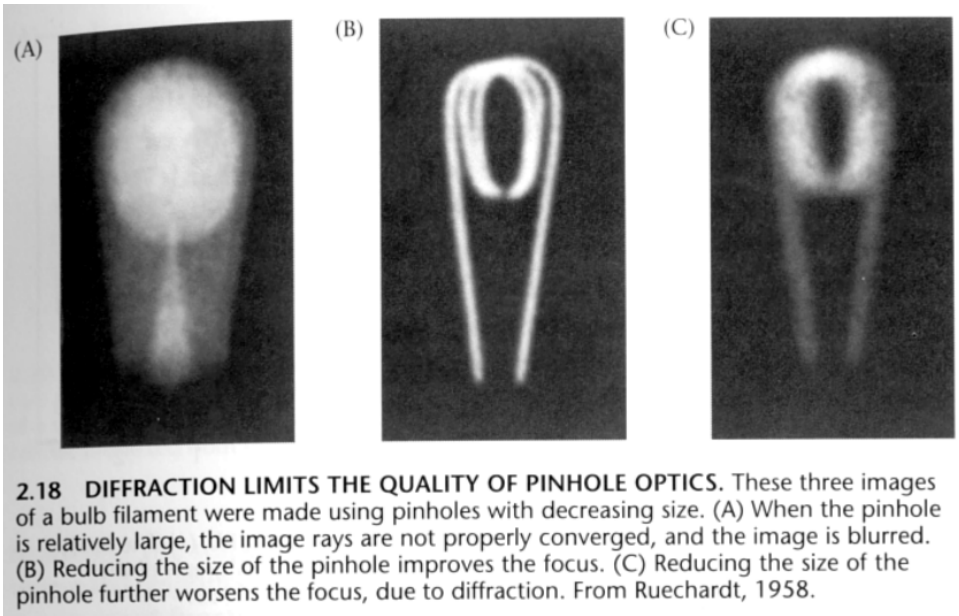
- **Effect of pinhole Size**



https://www.flickr.com/photos/dominicspics/4589827160

Pinhole Camera

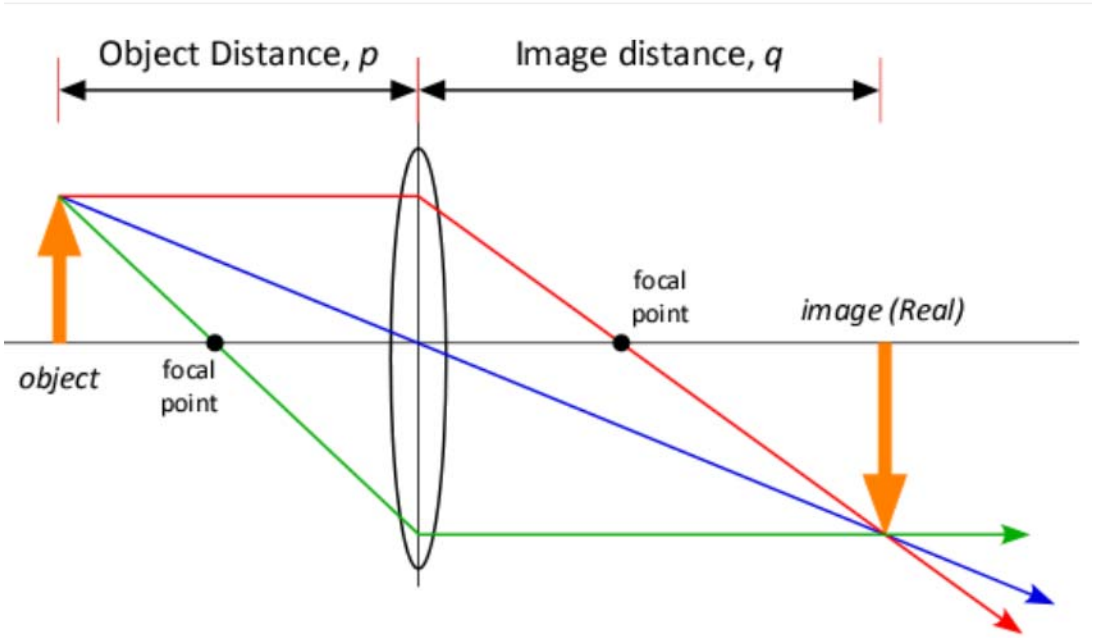
- Why not making the aperture as small as possible?
  - Less light
  - Diffraction effects



2.18 DIFFRACTION LIMITS THE QUALITY OF PINHOLE OPTICS. These three images of a bulb filament were made using pinholes with decreasing size. (A) When the pinhole is relatively large, the image rays are not properly converged, and the image is blurred. (B) Reducing the size of the pinhole improves the focus. (C) Reducing the size of the pinhole further worsens the focus, due to diffraction. From Ruechardt, 1958.

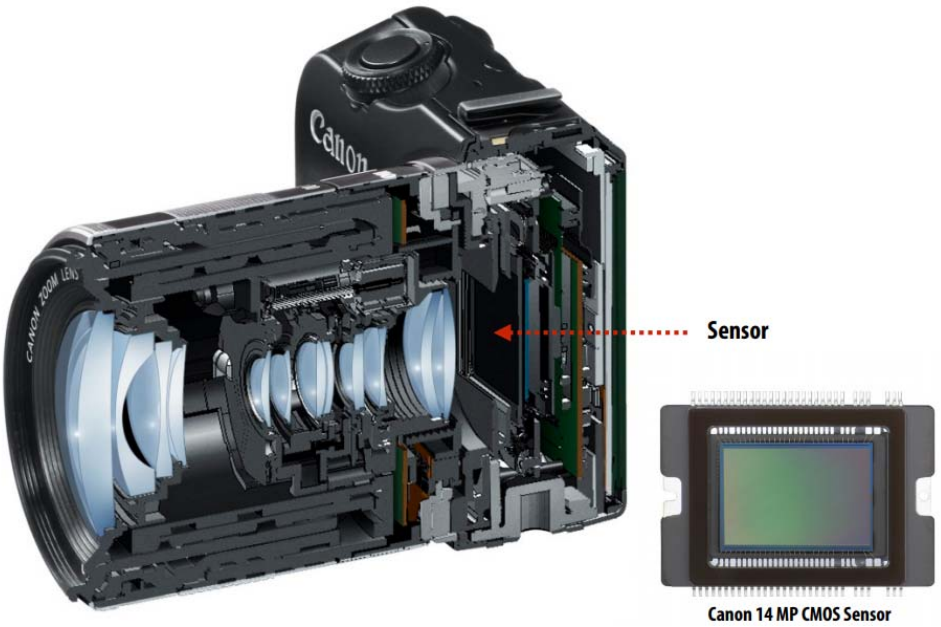
Wandell, Foundations of vision, Sinauer, 1995
http://6.869.csail.mit.edu/fa13/lectures/lecture10imageformation.pdf

Lenz camera

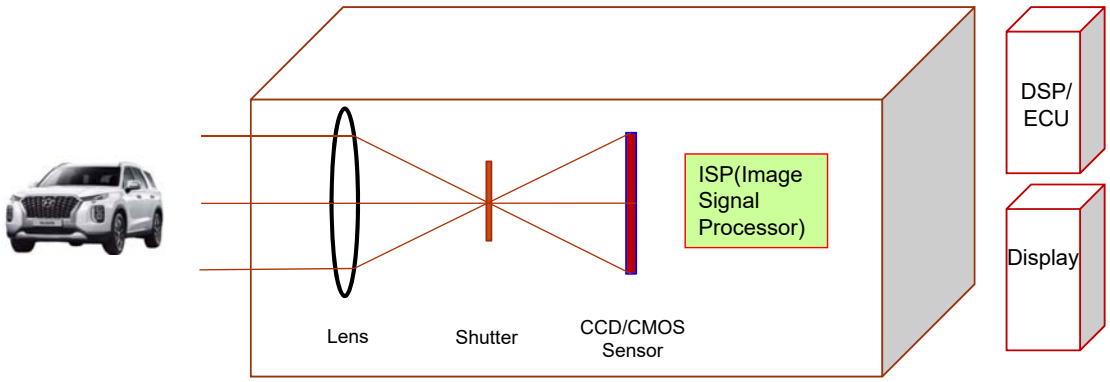


https://physics.stackexchange.com/questions/464866/determining-focal-length-of-converging-lens

Digital Camera

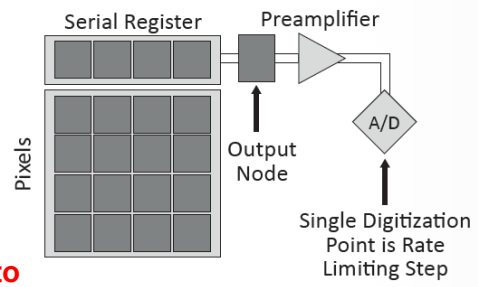


Digital Camera Structure



Camera sensor

- **Image sensors for camera**
  - Charge Coupled Devices(CCD)
  - Complementary Metal Oxide Semiconductors(CMOS)
- **CCD (Global Shutter)**
  - Every pixel is exposed simultaneously at the same instant in time
  - The signal from each pixel is **serially transferred to a single A/D C**
- **CMOS (Rolling Shutter)**
  - Eliminate the bottleneck by **using an A/D for every column of pixels**
  - The entire sensor array must still be converted one row at a time
  - Each individual row is typically able to begin the next frame's exposure



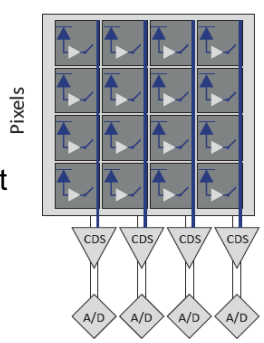
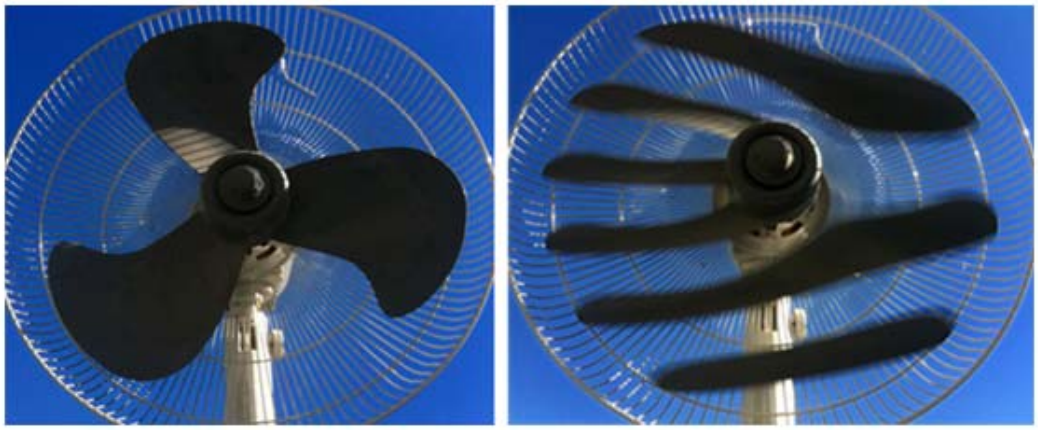


Figure 1. Basic schematics of CCD (above) and CMOS (left) sensor architectures.

Rolling shutter effects

- Rolling shutter effect : spatial distortions

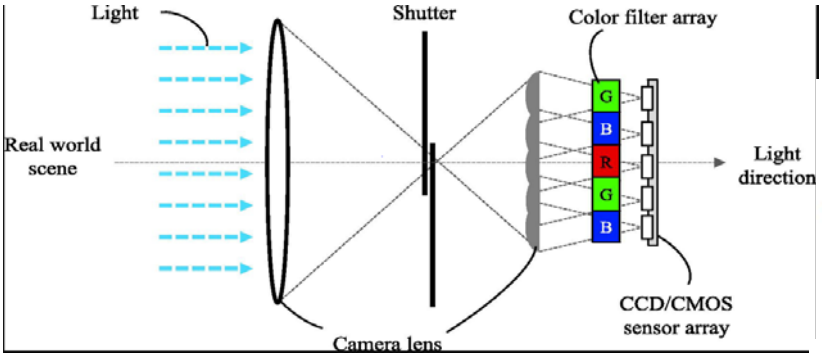


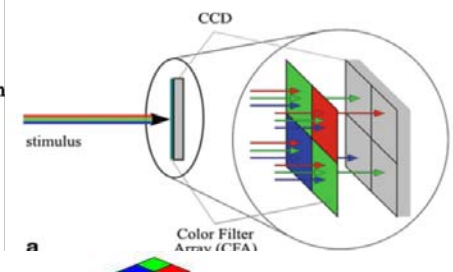
Motion artifacts in rolling shutter camera with moving objects

[*Reference*] http://lfa.mobivap.uva.es/~fradelg/phd/notes/global-shutter.html

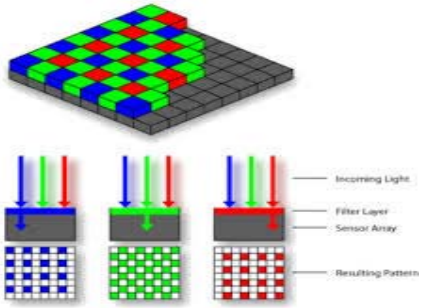
https://www.youtube.com/watch?v=dNVtMmLlnoE

Color Filter Array(CFA)/ Bayer layer









Bayer filter images

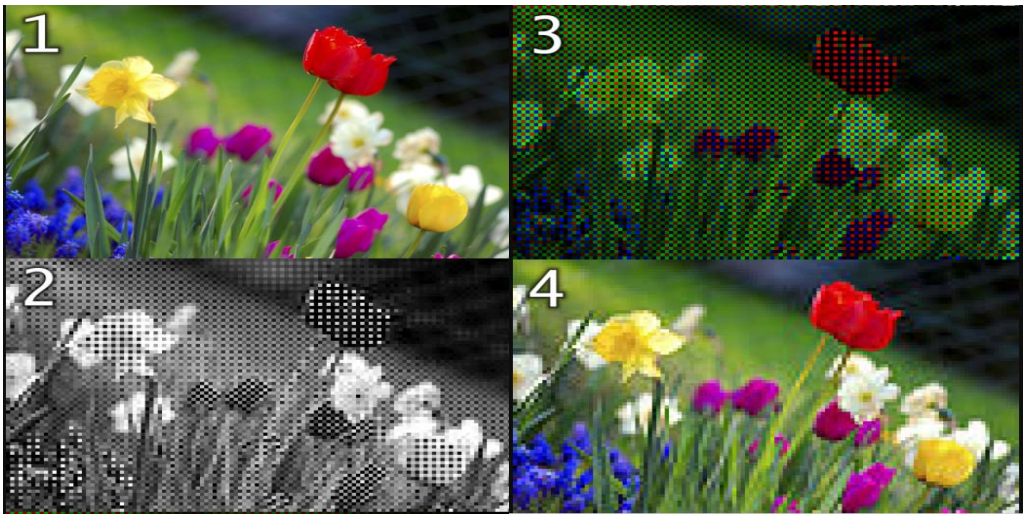






https://en.wikipedia.org/wiki/Demosaicing

Bayer filter images ➔ reconstruction

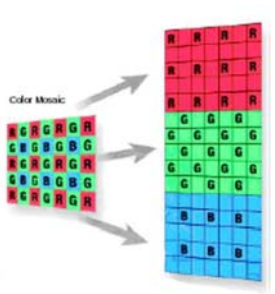


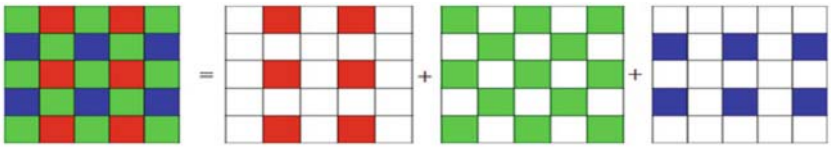
1. Original scene
2. Output of a 120×80-pixel sensor with a Bayer filter

3. Output color-coded with Bayer filter colors
4. Reconstructed image after interpolating missing color information

https://en.wikipedia.org/wiki/Demosaicing

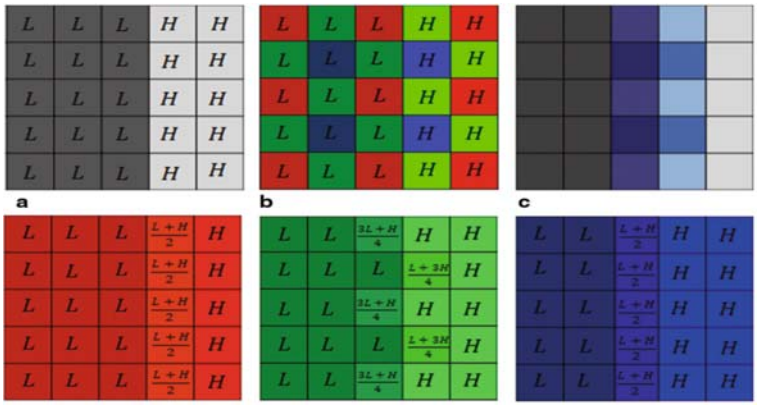
De-mosaic operation: Interpolation





$$K_G = \frac{1}{4}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$K_B = K_R = \frac{1}{4}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

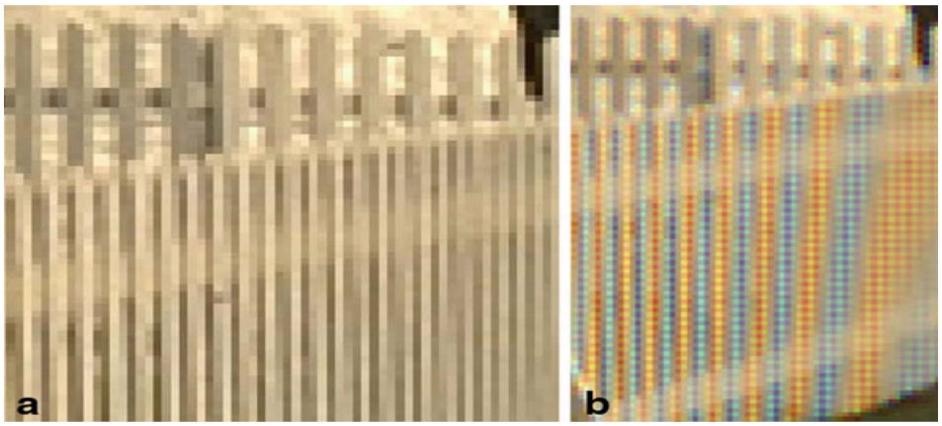


Artifacts of Bilinear Interpolation

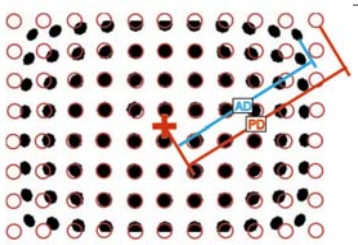


Artifacts of Bilinear Interpolation

- Zipper effect: refers to the abrupt or unnatural changes of intensities over a number of neighboring pixels, manifesting as an “on–off” pattern in regions around edges
- False color: appears are spurious colors which are not present in the original image, appearing as sudden hue changes due to inconsistency among the three color planes.



Lens Distortion Correction

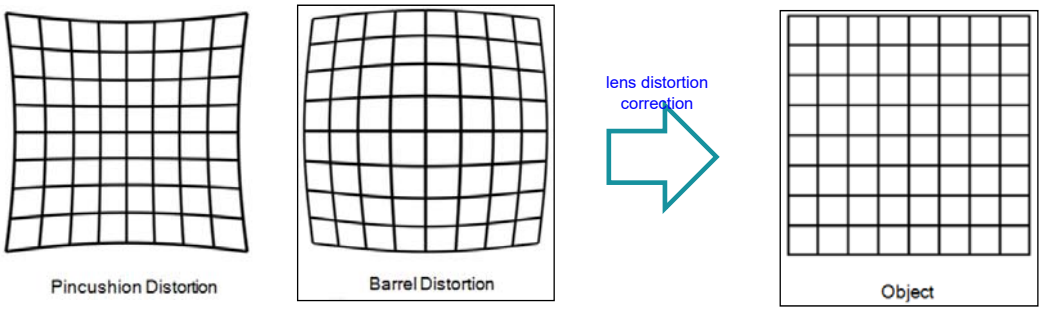


◆ Purpose

To generate warping table for lens distortion correction by using radial distortion property of lens

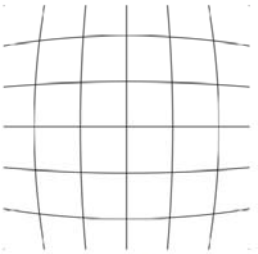
◆ Processes

① Measure radial distortion parameter by camera manufacture
② Generate distorted coordinates and coordinates from radial distortion by using equation in PC
③ Apply distorted coordinates and coordinates in warping table
④ Apply interpolation



Lens Distortion Correction

■ Lens distortion



Barrel Distortion

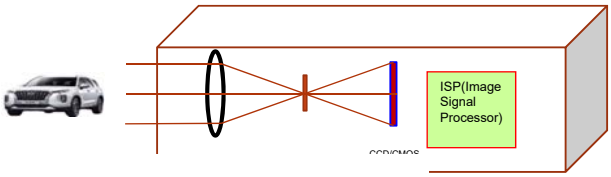


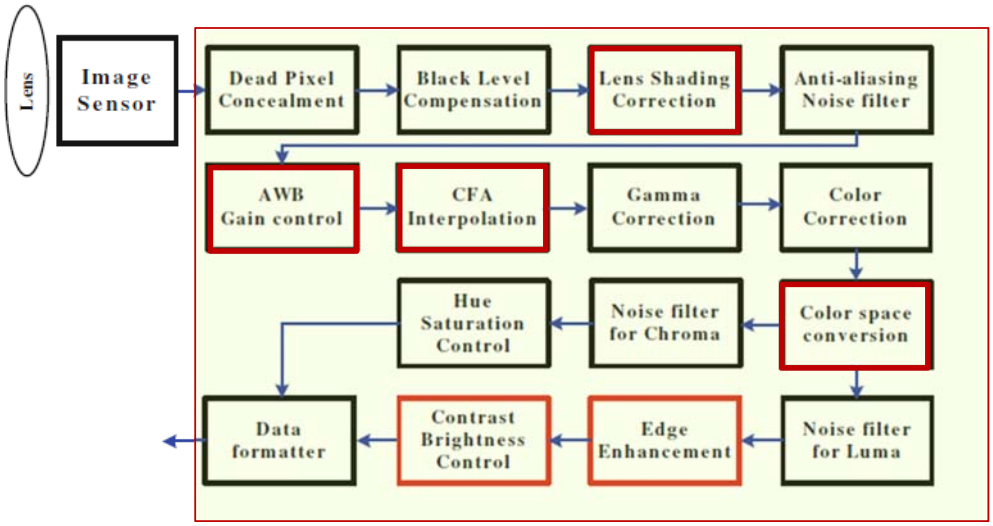
Captured Image



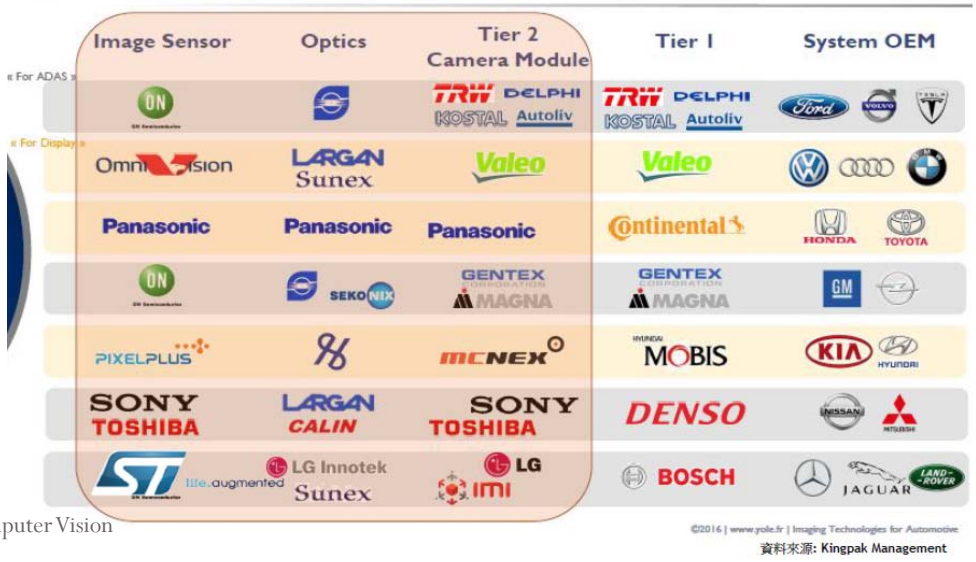
Corrected Image

ISP(Image Signal Processor)





Thank You
Next : Camera Sensor



Computer Vision
Report 03: PSNR 구하기

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Dept. of Information and Communication
Ho-Youl JUNG

영남대학교 Yeungnam University

주요내용

- Report 02 에서 RGB (원본)➔ YCbCr ➔ RGB(복원) 컬러 영상(재생)으로 변환
- RGB (원본) 과 RGB 컬러 영상(복원) 의 PSNR 구하는 프로그램 작성

Image quality measurement

- MSE(Mean Square Error) for Grayscale Images, $I(i,j)$ and $R(i,j)$.

$$MSE = \frac{1}{M \times N} \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} (I(i,j) - R(i,j))^2$$

- MSE(Mean Square Error) for Color Images, $I(i,j,c)$ and $R(i,j,c)$.

$$MSE = \frac{1}{3 \times M \times N} \sum_{c=1}^{3} \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} (I(i,j,c) - R(i,j,c))^2$$

- PSNR(Peak to Signal-to-Noise Ratio)

$$PSNR = 10 \times \log_{10}\left(\frac{MAX^2}{MSE}\right) = 20 \times \log_{10}\left(\frac{MAX}{\sqrt{MSE}}\right)$$

- Where MAX is maximum value of pixel, 255.

Report #3

- Report #2에서 작성한 프로그램에서 RGB (원본) 과 RGB 칼라 영상(복원) 의 PSNR 함수(getPSNR())을 비교한다.
- 구현한 프로그램의 코드와 실행 결과를 워드 또는 아래 한글 파일에 첨부하여 제출

- 파일명 : report03_성명_학번.hwp

- 2 가지로 구현 하여 그 결과를 비교
  - 구현 1 : OpenCV PSNR 함수를 이용하여 구현
  - 구현 2 : 앞에서 주어진 수식을 이용하여 각자의 PSNR 구하는 함수를 직접 구현
  - 구현 3 : time 모듈을 사용하여 PSNR OpenCV 함수와 직접 구현한 함수의 실행시간 비교
  - 결과 PSNR이 동일한지 확인하여 그 프로그램과 결과물을 제출
- Python 또는 C++ 중 하나만 구현

Report #3 실행 결과 예)

